

Stagehand Phase Evidence

Current checkpoint: public Site v64, verified 2026-09-03.

Project

Stagehand is a WebMCP-native 2D animation studio for editable paper-cutout scenes. The current cut supports rigged characters, imported image props, camera work, captions, non-voice audio cues, storyboard beats, multi-scene Preview, and deterministic WebM export.

Private source: github.com/swaymun/stagehand-animation-studio

Commit: 25731eaa9f8ad850e30ae9fc724135f95fe3f34d

Public Site: stagehand-animation-studio.saimun-h-shahee.chatgpt.site

Sites version: 64

Acceptance loop

Implement -> local tests -> Sites deployment -> hosted test -> UI roast -> fixes -> second review.

Implemented at this checkpoint

- Structured PropKeyframe data with interpolated X, Y, scale, and rotation.
- WebMCP tools: `get_prop_keyframes`, `set_prop_keyframe`, and `apply_prop_preset`.
- Human Assets controls: Pop in, Nudge, and X/Y/Scale/Rot at the playhead.
- Prop state flows through persistence, validation, thumbnails, Preview, scene operations, and WebM rendering.
- Preview hides the inspector, scene tools, duration editing, and mutation actions while retaining transport and Exit preview.
- Timeline keyframe buttons use 22 by 22 pixel hit targets while preserving the small diamond visual.

Verification

Local command gate:

```
npm run format
npm run lint
npm run build
npm run smoke
```

Local smoke passed with 50 registered tools, no page errors, human prop X edit 63.0, agent prop keyframe and preset, undo/redo, a second scene, and a two-scene WebM result of 1000 ms with downloaded bytes greater than zero.

Hosted command gate:

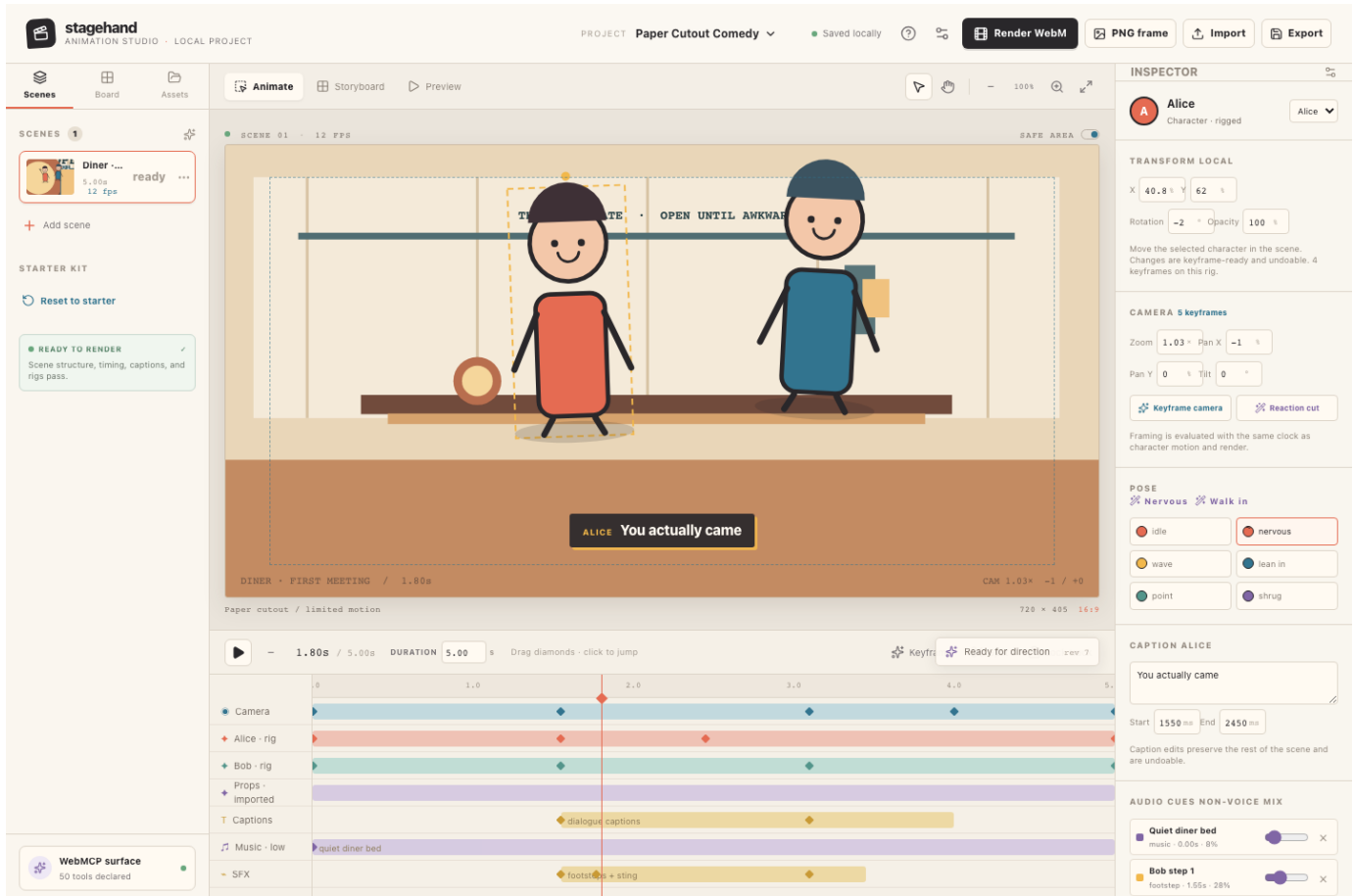
```
STAGEHAND_URL=https://stagehand-animation-studio.saimun-h-shahee.chatgpt.site npm run smoke
```

Hosted Playwright fallback passed the same prop workflow, two-scene WebM download, Preview state, and zero page errors. This is not proof of live production ChatGPT WebMCP discovery.

UI review evidence

Screenshots below were captured from the hosted v63 build at 1440 by 960. The fixture prop is intentionally a tiny black PNG used only to prove import, transform, and render plumbing.

Animate editing workspace



Assets panel with imported prop controls

The screenshot displays the Stagehand Animation Studio interface for a project titled "Paper Cutout Comedy". The interface is divided into several panels:

- Assets Panel (Left):** Lists various assets including "Alice - rigged", "Bob - rigged", "Diner background", "Coffee mug", and "evidence-prop". Each asset has a small thumbnail and a description. The "evidence-prop" asset is currently selected, showing its motion controls (Pop in, Nudge) and a playhead at 1.80s.
- Central Workspace:** Shows a scene titled "SCENE 01 - 12 FPS". It features two characters, Alice (red) and Bob (blue), in a diner setting. A text box at the bottom of the scene reads "ALICE You actually came". The scene is labeled "DINER - FIRST MEETING / 1.80s".
- Inspector Panel (Right):** Provides detailed controls for the selected asset, Alice. It includes:
 - TRANSFORM LOCAL:** X: 40.8, Y: 62, Rotation: -2, Opacity: 100.
 - CAMERA 5 keyframes:** Zoom: 1.03, Pan X: -1, Pan Y: 0, Tilt: 0.
 - POSE:** Nervous (selected), Walk in.
 - CAPTION ALICE:** "You actually came" with start and end times (1550 ms to 2450 ms).
 - AUDIO CUES NON-VOICE MIX:** Quiet diner bed (0.00s - 8%), Bob step 1 (1.55s - 28%).
- Timeline (Bottom):** A horizontal timeline showing the duration of the scene (5.00s). It includes tracks for Camera, Alice - rig, Bob - rig, Props - imported, Captions, Music - low, and SFX. A playhead is positioned at 1.80s.

Review-first Preview player

stagehand

ANIMATION STUDIO - LOCAL PROJECT

PROJECT Paper Cutout Comedy

Saving...

Render WebM

PNG frame

Import

Export

Scenes

Board

Assets

ASSETS 5

Alice - rigged

Rigged character - starter

Warm coral paper protagonist; keep her

Bob - rigged

Rigged character - starter

Diner tool foil character; enters from

Diner background

Background - starter

Warm late-night diner backdrop; leave the

Coffee mug

Prop - starter

Small amber prop for the coffee-spill

evidence-prop

Prop - imported

MOTION Pop in Nudge

AT PLAYHEAD X 68.1 % Y 56.1 Scale 1 X Rot 0

Story prop with a simple silhouette and

ADD PLACEHOLDER

Rigged character

Background

Prop

Audio

import character art

import pose sheet - auto

import background

import prop

WebMCP surface

50 tools declared

Preview

Exit preview

PREVIEW PLAYBACK Playing Scene 01 — Diner · first meeting Preview uses the same deterministic clock as render.

SCENE 01 12 FPS

SAFE AREA

ALICE You actually came

DINER · FIRST MEETING / 2.05s

CAR 1.05x -1 / +0

Paper cutout / limited motion 720 x 485 16:9

2.05s / 5.00s

Human · Apply pop-in to evidence-prop rev 9

Camera

Alice · rig

Bob · rig

Props · imported

Captions

Music · low

SFX

dialogue captions

quiet diner bed

footstep · sting

Roast findings and fixes

- **Imported props lacked direct human transform editing.** Added X/Y/Scale/Rot controls sharing the keyframe command path.
- **Four prop fields initially truncated.** Overrode the inherited asset-list flex rule with a readable two-column grid.
- **Preview carried too much editing chrome.** Converted it into a review-first player with an Exit preview affordance.
- **Timeline diamonds had tiny hit targets.** Enlarged the buttons to 22 by 22 pixels without changing the diamond glyph.

Limits

Direct CUA inspection was unavailable because the Mac was locked; Playwright was used for hosted interaction and screenshots. The smoke harness proves injected modelContext registration and execution, not live production ChatGPT WebMCP discovery. TTS, voice cloning, dialogue recording, phoneme extraction, and lip-sync remain out of scope.